

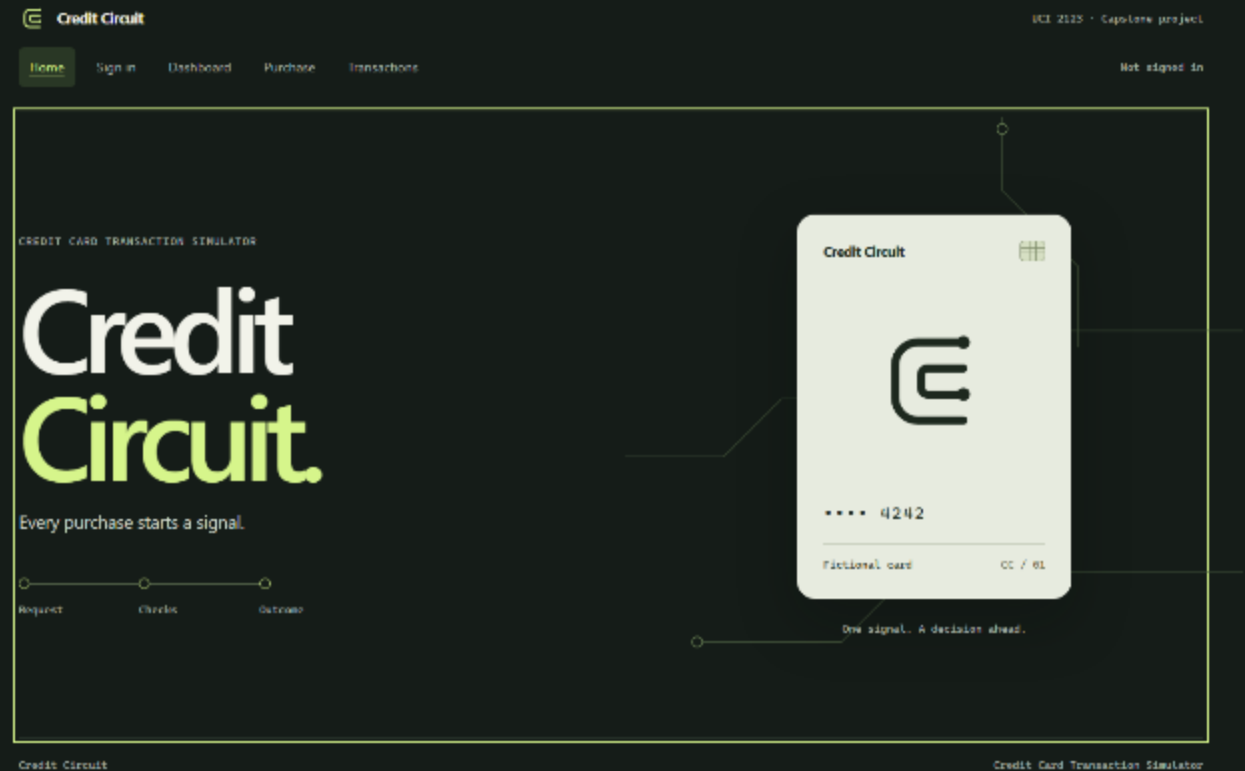
Credit Circuit

Fictional purchases with clear outcomes

UCI 2123 capstone

October 13, 2026

Local React, Spring Boot, MySQL



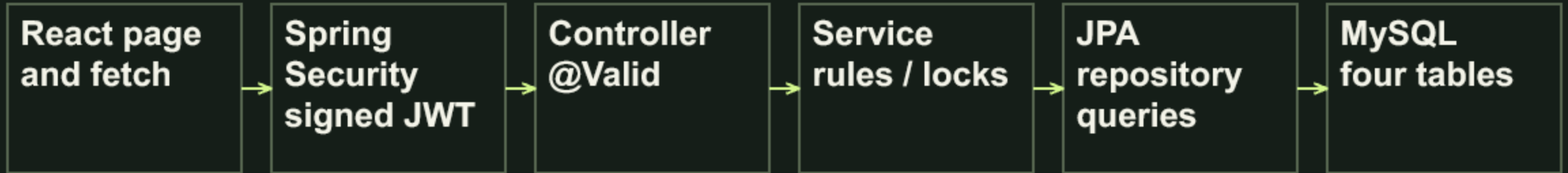
The problem

An uncertain response needs a safe retry.

A customer must reach only their own account.
A balance change must agree with its history.
A decline needs a clear financial explanation.

Approved local scope: application and quality evidence.
AWS and related deployment work are waived.

Application architecture



Boundary validation checks format.
Services enforce ownership and money rules.

One API, one database, one JWT authentication path.

Four tables

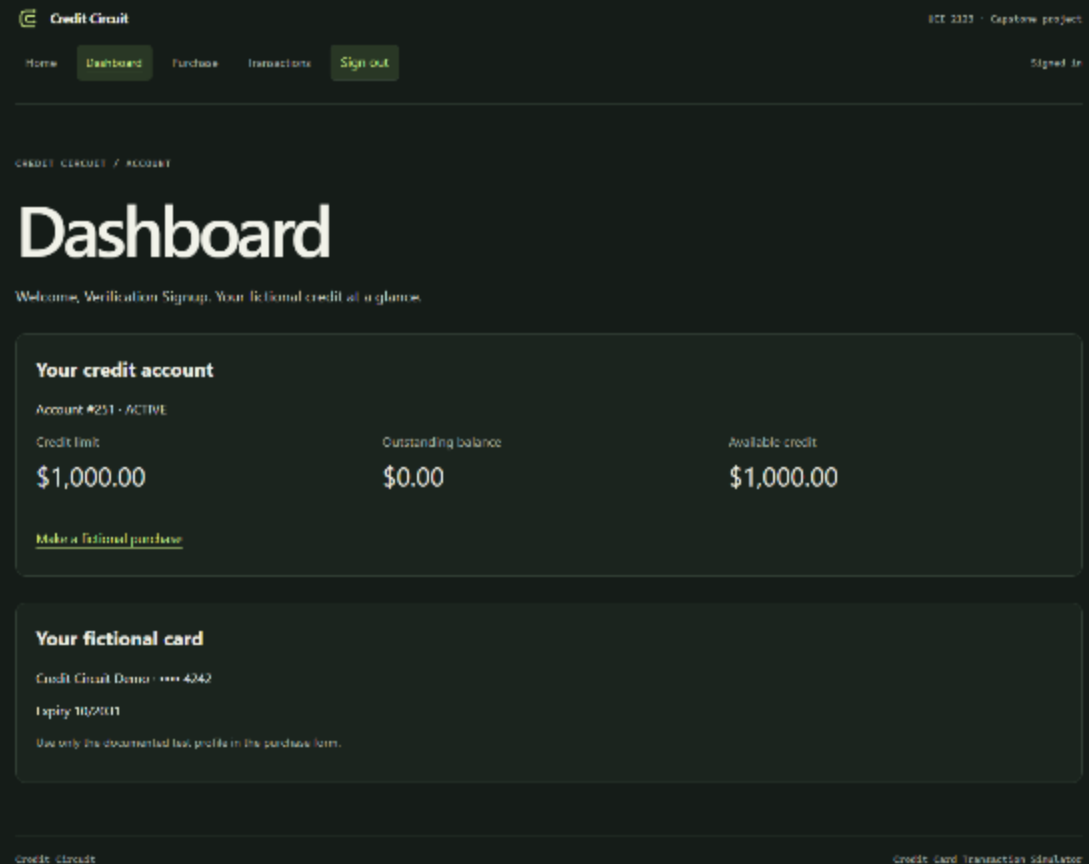
Table	Purpose
app_users	Identity, BCrypt hash, role
credit_accounts	Limit, outstanding balance, ACTIVE/FROZEN
demo_cards	Profile, masked display, expiry
card_transactions	Purchase/refund outcomes and request IDs

Available credit = limit – outstanding balance

A refund links to its original purchase once.

Live customer and admin demo

Register and sign in
Purchase \$50
Save a \$2,000 decline
Refund the original \$50
Freeze and reactivate



Every value in the demonstration is fictional.

Retries and consistent balances

Protection	Job
Stable request UUID	An identical retry returns the saved transaction
Account write lock	Competing balance changes wait their turn
Database transaction	Balance and history commit or roll back together

201: new financial outcome

200: identical saved retry

DECLINED: saved decision with no balance increase

Authentication and access

BCrypt password checks and signed JWT verification
USER/ADMIN routes plus account ownership
Access stays in React memory for 15 minutes
Local CORS and authentication rate limiting

Sign-out discards browser access.

A copied token remains valid until expiration.

Verification evidence

Check	Measured result
JUnit and MySQL	36 passed, including 10 MySQL tests
JaCoCo Java lines	95.74%, with a 70% minimum gate
Postman	49 requests, 72 assertions passed
Chromium workflows	11 groups passed
SonarQube	Actual Java/React analysis and saved report

Reports distinguish real behavior from injected failure cases.

Lessons and remaining work

Retries need stable identity.

Locks and transactions solve different problems.

Quality claims need actual evidence.

Planned 3D card after required application work

Two peer reviews and recorded changes

Rehearsal October 12

Presentation and submission October 13